

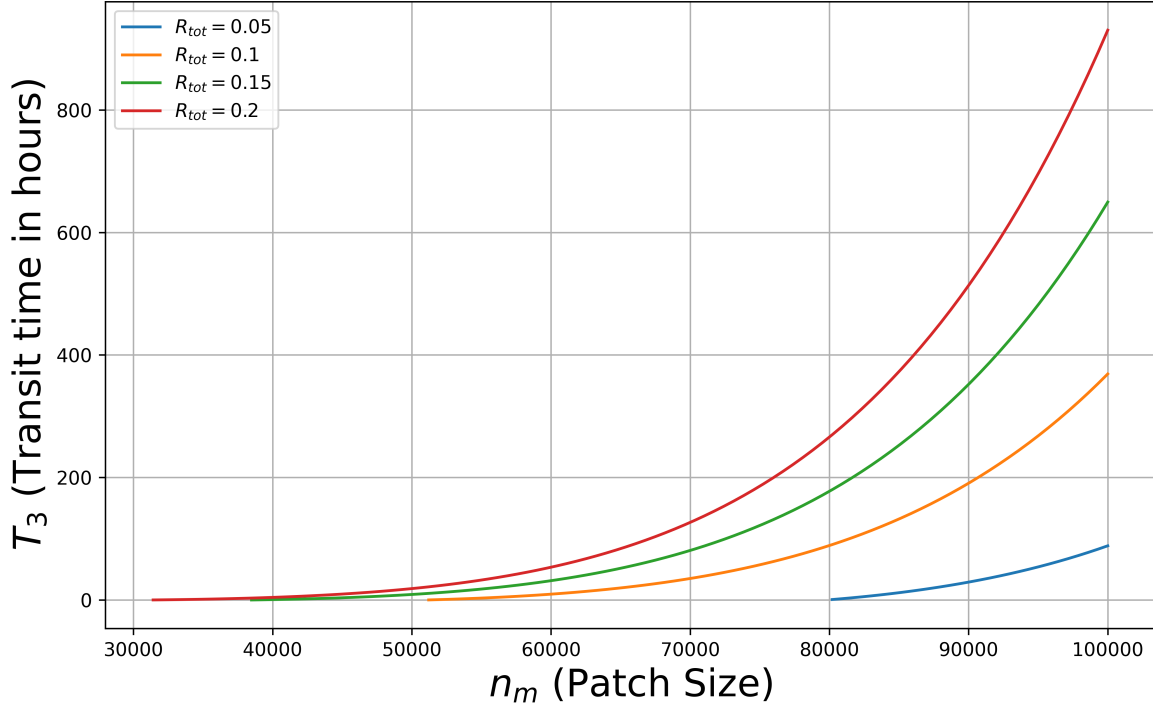


Figure 4. Transport time T_3 plotted against qLDPC patch size n_m , for various tolerable LFRs. The analysis demonstrates a positive correlation between patch size and the maximum tolerable transport time T_3 . This relationship stems from the enhanced logical error tolerance of larger patches, which enables longer-distance transport. The plots exhibit a lower limit on patch size, at which the T_3 value becomes zero. This occurs when the sum of all error terms except R_3 exceeds the target error rate, regardless of T_3 . In such cases, the numerical solver attempts to compensate by assigning negative T_3 values, indicating that the target R_{tot} is unattainable for patches smaller than this threshold.

While it may be possible to extend this scheme to LP codes, such an extension would require additional assumptions about the ability to generate logical X operators at the edge of the patch. To maintain rigor and avoid unsubstantiated claims, we restrict our analysis to HGP codes, for which the teleportation scheme is well-defined. This decision ensures that our results are based on established protocols and avoids potential inaccuracies that could arise from extrapolating beyond the current state of knowledge in the field. Designing a teleportation protocol for these other non-HGP qLDPC code families can reduce patch sizes by a factor of two to four, reducing our resource requirements.

Another caveat is that of classically transferring measurement results onto the users' smartphones. In future research, if a protocol can be devised to teleport the logical qubits of a quantum error-correcting code patch onto a single photon, then photon measurement device can be built into smartphones, which makes the transfer quantum-secure. An alternate way, as outlined in the pseudocode 2 is to perform a heralded qubit transfer from each atomic physical qubit on the surface code to a photon [51, 52] and measure those photons on the

end-user's smartphone [31] in a pre-defined sequence and decode to extract the logical surface-code measurement, but this method has only 0.88 fidelity and 0.69 efficiency [51], which when propagated over all the atoms of a single surface code, would lead to high failure rates and high resource overheads. Therefore, development of a more efficient method of performing atom-to-photon qubit transfer would be beneficial towards a highly secure network with minimal resource overhead.

With this kind of network, one can easily imagine creating new symmetric keys between multiple users, for example, a consumer who wishes to secretly share her banking details with a large number of different vendors, all via guaranteed entanglement-generated quantum keys. We have so far considered a single Charlie Inc., with one long-term mass storage and qATMs distributed in the field. However, as long as Charlie Inc. has set up pre-existing and refreshed entanglement, there is no reason why it is not possible for Charlie Inc. to have multiple long-term quantum memory sites in geographically distinct locations, which we term quantum hubs. With a constant refresh of entanglement between Charlie Inc.'s hubs, in principle, global scale links could be achieved

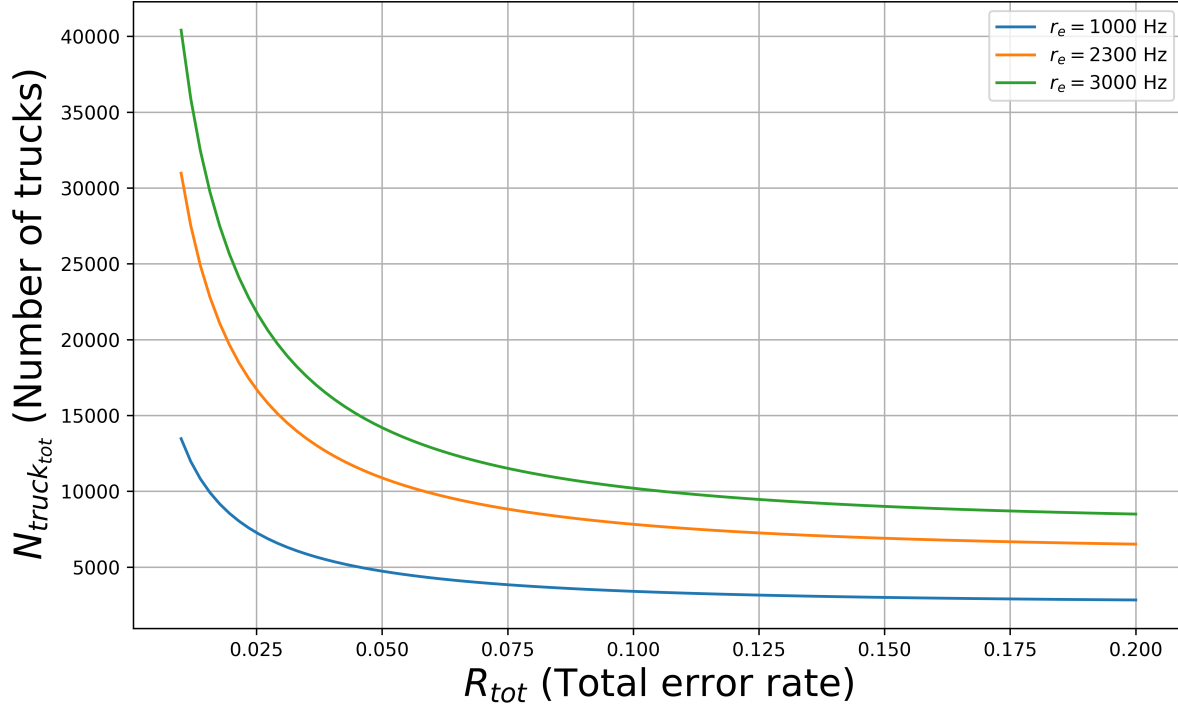


Figure 5. Number $N_{truck_{tot}}$ of trucks or train cars required for the network, as a function of overall LFR R_{tot} , showing an inverse relationship between the network’s tolerable error rate R_{tot} and the required number of transport vehicles (trucks or train cars) for a fixed transit duration of 90 minutes. As the acceptable R_{tot} increases, the fidelity requirement decreases, and smaller patch sizes become viable. These reduced patch sizes correspond to lower code distances, necessitating fewer error correction rounds. Consequently, the unloading process becomes faster, reducing the number of parallel qLDPC patches needed. This cascade effect ultimately results in a decreased demand for transport vehicles to maintain the desired bandwidth.

with additional entanglement-swapping operations.

-
- [1] J. P. Dowling and G. J. Milburn, Quantum technology: the second quantum revolution, Philosophical Transactions of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences **361**, 1655 (2003).
 - [2] I. H. Deutsch, Harnessing the power of the second quantum revolution, PRX Quantum **1**, 020101 (2020).
 - [3] L. Jaeger, The second quantum revolution, From Entanglement to Quantum Computing and Other Super-Technologies. Copernicus (2018).
 - [4] M. Atzori and R. Sessoli, The second quantum revolution: role and challenges of molecular chemistry, Journal of the American Chemical Society **141**, 11339 (2019).
 - [5] P. P. Rohde, *The quantum internet: The second quantum revolution* (Cambridge University Press, 2021).
 - [6] H. Riel, Quantum computing technology, in *2021 IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2021) pp. 1–3.
 - [7] J. Gambetta, IBM’s roadmap for scaling quantum technology, IBM Research Blog (September 2020) (2020).
 - [8] C. Gardiner and P. Zoller, *The quantum world of ultra-cold atoms and light book II: the physics of quantum-optical devices*, Vol. 4 (World Scientific Publishing Company, 2015).
 - [9] J. L. O’Brien, Optical quantum computing, Science **318**, 1567 (2007).
 - [10] M. Körber, O. Morin, S. Langenfeld, A. Neuzner, S. Ritter, and G. Rempe, Decoherence-protected memory for a single-photon qubit, Nature Photonics **12**, 18 (2018).
 - [11] S. Perseguers, G. Lapeyre, D. Cavalcanti, M. Lewenstein, and A. Acín, Distribution of entanglement in large-scale quantum networks, Reports on Progress in Physics **76**, 096001 (2013).
 - [12] P. P. Rohde, *The quantum internet: The second quantum revolution* (Cambridge University Press, 2021).
 - [13] M. A. Nielsen and I. L. Chuang, Quantum computation and quantum information, Phys. Today **54**, 60 (2001).
 - [14] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, Quantum entanglement, Reviews of modern physics **81**, 865 (2009).

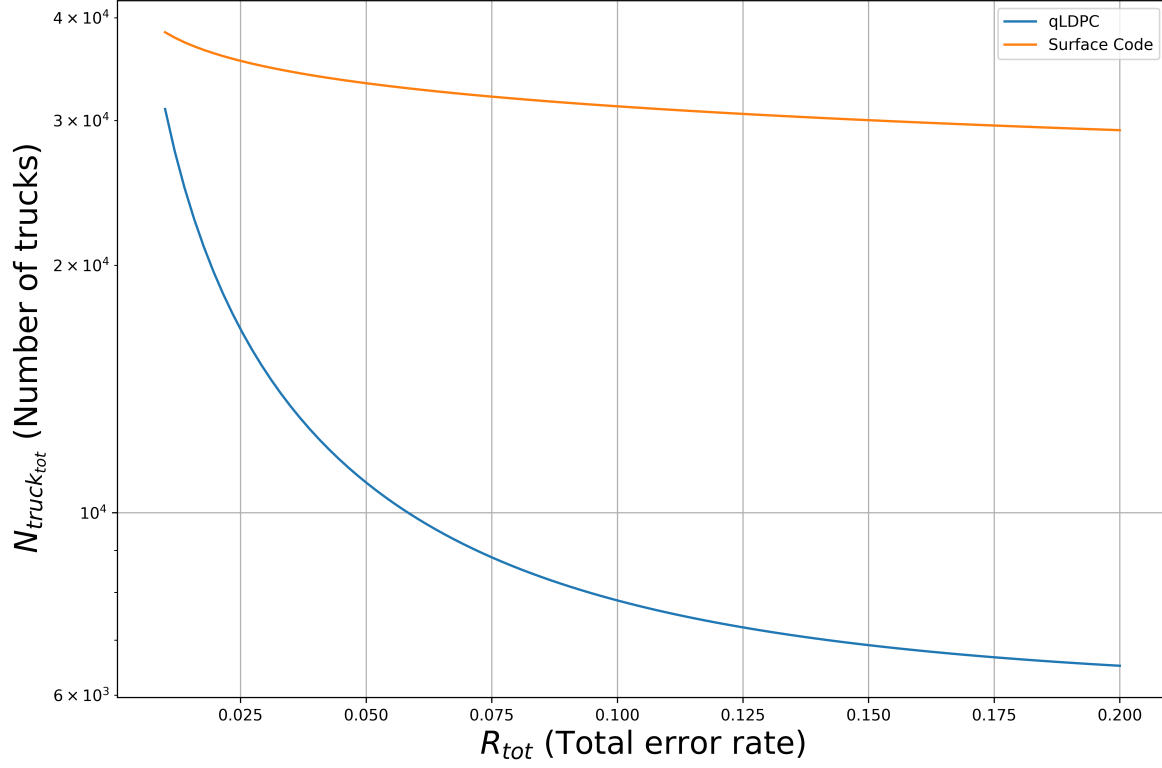


Figure 6. Resource Requirements (Number of vehicles) for Surface code vs qLDPC. The qLDPC network protocol demonstrates significantly higher efficiency in transport resource utilization compared to the surface code protocol, nearly by an order of magnitude.

- [15] S. Wehner, D. Elkouss, and R. Hanson, Quantum internet: A vision for the road ahead, *Science* **362**, eaam9288 (2018).
- [16] H. J. Kimble, The quantum internet, *Nature* **453**, 1023 (2008).
- [17] A. S. Cacciapuoti, M. Caleffi, F. Tafuri, F. S. Cataliotti, S. Gherardini, and G. Bianchi, Quantum internet: Networking challenges in distributed quantum computing, *IEEE Network* **34**, 137 (2019).
- [18] A. Sharma and A. Kumar, A survey on quantum key distribution, in *2019 International Conference on Issues and Challenges in Intelligent Computing Techniques (ICICT)*, Vol. 1 (IEEE, 2019) pp. 1–4.
- [19] A. Peres, Delayed choice for entanglement swapping, *Journal of Modern Optics* **47**, 139 (2000).
- [20] A. K. Ekert, Quantum cryptography based on bell’s theorem, *Physical review letters* **67**, 661 (1991).
- [21] S. J. Devitt, A. D. Greentree, A. M. Stephens, and R. Van Meter, High-speed quantum networking by ship, *Scientific reports* **6**, 36163 (2016).
- [22] M. Zhong, M. P. Hedges, R. L. Ahlefeldt, J. G. Bartholomew, S. E. Beavan, S. M. Wittig, J. J. Longdell, and M. J. Sellars, Optically addressable nuclear spins in a solid with a six-hour coherence time, *Nature* **517**, 177 (2015).
- [23] J.-P. Tillich and G. Zémor, Quantum ldpc codes with positive rate and minimum distance proportional to the square root of the blocklength, *IEEE Transactions on Information Theory* **60**, 1193 (2013).
- [24] N. P. Breuckmann and J. N. Eberhardt, Quantum low-density parity-check codes, *PRX Quantum* **2**, 040101 (2021).
- [25] D. A. Lidar and T. A. Brun, *Quantum error correction* (Cambridge university press, 2013).
- [26] L. Henriet, L. Beguin, A. Signoles, T. Lahaye, A. Browaeys, G.-O. Reymond, and C. Jurczak, Quantum computing with neutral atoms, *Quantum* **4**, 327 (2020).
- [27] Q. Xu, J. P. Bonilla Ataides, C. A. Pattison, N. Raveendran, D. Bluvstein, J. Wurtz, B. Vasić, M. D. Lukin, L. Jiang, and H. Zhou, Constant-overhead fault-tolerant quantum computation with reconfigurable atom arrays, *Nature Physics* , 1 (2024).
- [28] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, Surface codes: Towards practical large-scale quantum computation, *Physical Review A—Atomic, Molecular, and Optical Physics* **86**, 032324 (2012).
- [29] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, *Nature* **627**, 778 (2024).